Recommendations to Optimize Health in Youth Runners

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ABSTRACT

Youth running is defined as participation below age 18. Jumping/multidirectional loading sports (soccer, basketball) may optimize bone health. Psychological development requires monitoring to reduce risk of injuries and burnout. Adequate energy availability is essential for health; screening for disordered eating and optimizing nutrition is important. Injuries during growth spurts are best addressed by identifying the physical maturity of the runner and conventional treatment. Appropriate start of competition and competition distance should be individualized rather than using age-based recommendations and requires careful monitoring. Promoting foot strength and reducing landing impacts may lower injury risk.

INTRODUCTION

Participation in youth running is increasing (23,24). Youth runners (defined as runners ages 17 years and younger) have different needs than adult runners to promote health and longevity in the sport. In youth runners, we define health as normal and appropriate age- and maturity-specific physical and psychological development while participating in the sport of running (cross-country, track, and road or trail racing). This does not refer to land-based sports where running is part of sports participation (e.g., basketball and soccer). A study of running-related injuries presenting to U.S. emergency departments in 2007 compared with 1994 reported an increase of 34%, with the highest injury rates occurring in runners ages 12–14 years (45.8 per 100,000 U.S. population) (31). Youth runners require strategies to optimize training, musculoskeletal health, nutrition, psychological well-being, and running distances to reduce risk of injuries, overtraining, and burnout. We provide recommendations for common questions related to youth running based on current available science.

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WHAT ARE THE BEST ACTIVITIES TO RECOMMEND FOR YOUTH RUNNERS TO SUPPORT BONE DEVELOPMENT AND TO REDUCE FUTURE BONE STRESS INJURY?

Bone is a dynamic structure with strength increasing in response to hormones, nutrition, behavioral, and mechanical loads (43). The most critical time for bone development is during the 2 years surrounding peak height velocity when 26% of adult bone mass is acquired (29). Sports participation during time of peak bone mass accrual may improve bone architecture by increasing density and enhancing structural geometric properties (52,54). Youth athletes in sports that involve high-impact loading or multidirectional loading may have higher bone mineral density (BMD) and enhanced bone geometry at the anatomical sites specific to the loading demands of the given activity (56). By contrast, the repetitive impact loading from running alone does not provide predictable gains in bone health. Sports that do not include impact loading (such as swimming) may result in weaker bones compared with nonathletes (15).

Young runners who participate in higher impact, multidirectional sports have crossover benefits for

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bone health. Adolescent male runners who also participated in basketball had an 82% reduction in stress fracture risk (57). Older elite runners ages 18–44 years who participated in ball sports during their youth had a 50% lower risk of stress fracture during running compared with peers who had not (16). However, women with menstrual irregularities did not attain the same bone benefits from ball sports (16), supporting a model that accounts for multiple contributors influencing the response to skeletal loading.

Ground reaction forces >3.5 fold body weight with peak force <0.1 seconds may be the most effective stimuli for bone development (18). If young athletes are not exposed to these forces in their sports activity, they can cross-train with a jumping program that has high ground reaction forces delivered at a rapid rate. For example, adolescent cyclists and swimmers participating in a 9-month regular jumping program had significant improvement in both bone mineral content and bone geometry (58).

Recommendations. All young athletes should participate in high or multidirectional impact sports during childhood for at least 2 years. Young runners who do not participate in high impact or multidirectional sports should be in a structured jumping program to achieve gains in bone mass.

WHAT STRATEGIES CAN BE USED FOR YOUTH RUNNERS TO ASSESS PSYCHOLOGICAL DEVELOPMENT FOR SAFE PARTICIPATION AND TO AVOID BURNOUT?

Creating a safe and supportive environment for youth runners may promote longevity in the sport, increase overall physical activity, and contribute to psychological well-being. Youth runners are developing physically and psychologically while participating in the sport. Beyond age, other factors must be considered when assessing the psychological status of a youth runner. The source of motivation for the young runner should guide participation. A self-motivated young runner is more likely to succeed as a lifelong runner. When running behavior is directed or controlled by external rewards and punishment from coaches and/or family, perfectionism, training distress, and increased burnout are expected outcomes (5,30). External influence from parents and/or coaches is predictive of perfectionistic thoughts and increased stress; there are differences based on sex (5). Perfectionism has been associated with several psychological conditions, including eating disorders (53).

Perfectionism and coach or parent stressors can lead to burnout or overtraining in youth runners. Burnout in youth athletes may be defined as withdrawal from a previously enjoyable activity in response to chronic stress that was not mitigated by coping efforts (51). Sports specialization is more common in today's youth sport cultures, often defined as choosing a single main sport, training more than 8 months per year, and/or quitting all other sports to focus on a single sport (38). Highly specialized youth athletes have been noted to incur more injuries (41,44). Athletes commonly report that they discontinue a sport due to injury, burnout, time conflicts, and/or greater interest in other activities (13).

Recommendations. It is important to assess each youth runner's psychological development, motivation, and readiness to determine safe participation in youth running. Monitoring for recurrent injuries, imbalanced training, and/or inadequate recovery can help identify burnout. Youth sport specialization should not be encouraged, as this may not result in improved performance and often contributes to increased risk of discontinuation of the sport.

ARE THERE WAYS TO SCREEN FOR APPROPRIATE CALORIC INTAKE IN YOUTH RUNNERS?

Adequate energy intake (EI) in athletes is of particular concern in leanness sports including youth running. Energy availability (EA) (28) is the energy remaining for normal physiological functioning after that which has been utilized for exercise. EA has traditionally been defined as EI minus exercise energy expenditure, normalized to fat-free mass (FFM) (19,39). Studies in women suggest that EA should be at least 30 kcal/kg FFM per day, as EA below this value results in changes in markers of metabolic and endocrine function (28). Adequate EA of 45 kcal/kg FFM per day may be more appropriate, particularly in youth athletes, who also have energy requirements for growth and development (2). A specific cutoff has not been determined in male runners. Appropriate EA thresholds likely vary among individuals. Low EA can have significant health and performance consequences (36). However, determining appropriate EI and EA in athletes can be difficult. Dietary and training recall can be inaccurate, and direct measurement can be expensive and labor-intensive (20). Therefore, surrogate markers of low EA are important (1).

Low EA may result in poor health and impaired running performance. The interrelationship of low EA, menstrual dysfunction, and low BMD is known as the female athlete triad (Triad) (39). The International Olympic Committee has introduced a new term of relative energy deficiency in sport (RED-S) (34,36). RED-S describes health and performance consequences of low EA which may occur in both male and female athletes. In a study of 1,000 female adolescent and young adult athletes who completed surveys about their health and sports performance, positive responses to eating disorder/disordered eating survey questions were used as surrogates markers for low EA/RED-S; these positively correlated with health and performance decrements defined in the RED-S model (1). The RED-S model also identifies growth and development as being negatively impacted by low EA (34,36), so monitoring growth curves is important to ensure both needs for growth and sports participation are being addressed.

Low EA may occur with or without eating disorders/disordered eating. Screening for behaviors associated with eating disorders/disordered eating is recommended at the time of the Preparticipation Physical Evaluation (4,22), and appropriate nutrition should be monitored by coaches, strength and conditioning professionals, and others who interact with the youth athlete. While validated in older female athletes, the Low Energy Availability in Females Questionnaire was developed specifically to determine low EA and Triad risk in athletes and focuses largely on injuries and gastrointestinal and reproductive function (32). A Triad cumulative risk assessment (Triad-CRA) (12) and RED-S screening tool (RED-S CAT) (35) have both been developed. Although studies in youth runners are lacking, in college runners designated moderate- and high-risk Triad-CRA, 50% and 71% of runners sustained a bone stress injury (BSI) within a year, respectively (55). These findings are consistent with a previous report that suggested cumulative Triad risk factors increase the risk of stress fractures in adolescent female runners (57). In a similar study of male collegiate runners, higher modified Triad-CRA scores (excluding menstrual history) predicted increased rates of BSI (25). The RED-S CAT has been less tested than the Triad-CRA; however, in a study of 1,000 adolescent and young adult female athletes, there was considerable agreement between the 2 screening tools when assigning participants to moderate-/high-risk categories (21).

Recommendations. Youth runners of both sexes should be screened for low EA and health markers of Triad/RED-S using screening tools such as Triad-CRA and RED-S CAT. Their body mass index (BMI) and weight (percent median BMI and expected weight percentile) should be calculated, with careful attention to their individual growth curves. Detailed menstrual history should be obtained in all female runners, and male runners should be screened for markers of low testosterone (such as loss of morning erections). Youth runners in elevated risk categories should have a medical workup for menstrual dysfunction and impaired bone health and meet with a sports dietitian to improve EA. Early recognition and treatment of eating disorders/disordered eating is critical for lifelong health and longevity in sport.

ARE YOUTH RUNNERS AT AN INCREASED RISK OF INJURY DURING GROWTH SPURTS?

A unique consideration in an adolescent runner compared with an adult is physical maturation (8,23,24,27,37). Specifically, two important factors that may contribute to potential injuries are peak height velocity (PHV) and change in bone mineral content. On average, the age of PHV is 14 years for boys and 12 years for girls (27). Peak height velocity is a period of rapid growth when long bones change length at a relatively faster rate than the muscle-tendon-bone complex (8,24). The increase in bone mineral content deposition typically follows peak height velocity, occurring at age 16–18 years for boys and 12–14 years for girls (14,50). More importantly, bone mineral content is at its lowest level just before peak height velocity (14), and almost half of one's adult bone mineral content accrues during adolescence (6). It is theorized that the variable rate of change in bone, tendon, muscle, cartilage, and growth plate may affect running biomechanics and tolerance to load in the maturing youth runner, placing the youth runner at increased risk of injury (24). This potential for increased risk of injury during growth spurts requires careful monitoring of training load to reduce risk as the bones are actively growing (13).

Recommendations. An appropriate running training and competition program should take into account an individual's growth and development rather than chronological age.

WHEN SHOULD A YOUTH RUNNER START COMPETING AND WHAT DISTANCES ARE APPROPRIATE?

Evidence-based guidelines on appropriate race distances by age are not available for youth runners. Eligibility status for youth runners competing in endurance events, including half and full marathon distances, is controversial. Counseling young runners and parents is currently based on expert opinion and personal bias. Normal growth and development along with psychological status must be considered in the decision for safe race distance and pace.

Limited available reports suggest youth runners can safely complete longer distance races. Among 310 youth runners ages 7–17 years who completed the Twin Cities Marathon over 26 years, only 4 required finish area medical assistance and all were for minor medical issues (49). Even in the heat of the 2007 Twin Cities Marathon, none of the 20 runners younger than 18 years required medical assistance, while a record number of adult runners dropped out of the race or required medical care (48). From 2011 through 2017, an additional 660 runners under 18 completed the marathon with no apparent increase in medical encounters. In 2018, a 15-year-old runner placed sixth in the open women's division without visible harm (7). The Students Run LA program (1989–2018) has had more than 63,000 youth runners complete the marathon distance with no reports of adverse outcomes (10).

Recommendations. Self-motivated youth runners may complete a given race distance, administration rules permitting, with the following recommendations during training: complete a supervised training program, remain pain and injury free, continue appropriate weight and height gains, maintain health including nutrition and sleep, maintain good social interaction and academic performance, and girls should experience normal menstrual function (47).

While not evidence-based, longer running events (such as half and full marathon) should not keep youth age group time records below age 18 years and should not publicize the participation of child or adolescent runners. Longer events should be completed with an adult, and youth runners should avoid large events where it is easy to "get lost in the crowd." Nutrition and hydration are important to monitor in youth athletes during participation in long-distance races, and race starts shortly after sunrise as advised to avoid the heat of the day (47).

WHAT FOOTWEAR IS RECOMMENDED FOR YOUTH RUNNERS?

Humans have been running for approximately 2 million years, with the first evidence of footwear emerging 10,000 years ago (11). These early shoes were fabricated from sagebrush bark and comprised a flat sole for plantar protection with straps to keep it on the foot. The human foot is a complex structure with 26 bones and 33 articulations, each with 6 degrees of freedom of motion. The foot has more than 20 muscles, 10 of which are in the arch and provide support to the foot during landing. This complex flexible structure allows the foot to be highly adaptable, acting as a flexible structure to help attenuate load during weight acceptance, as well as a rigid lever for push-off. Therefore, the foot is naturally mobile and very well adapted to walk and run without footwear.

Modern shoes have become increasingly supportive, reducing the demand on the foot and reducing foot strength. Evidence to support this comes from populations that do not use modern footwear. A study of 2,300 Indian children revealed that habitually barefoot children demonstrate a low incidence of flat feet compared with habitually shod children (42). Another study of Kenyan children found that those who were habitually barefoot had stronger feet and significantly reduced lower extremity injury incidence (3). The added cushioning in modern footwear changes the landing mechanics of runners. Studies have shown that running in cushioned shoes encourages a heel strike landing as opposed to landing on the ball of the foot as habitual barefoot runners do (26). The heel strike landing results in an abrupt increase in forces transmitted (45), which has been related to overuse musculoskeletal injuries in adult runners (17,33,40).

Modern minimal shoes are designed to mimic barefoot gait, eliminating any support or cushioning. Studies of adults transitioning to minimal shoes in both walking and running have demonstrated increases in the size and strength of their arch muscles (9,46). In addition, those habituated to minimal shoes for running may be less likely to rearfoot strike, and this can promote softer landings.

Recommendations. Youth runners benefit from strategies to strengthen the foot and ankle. The use of minimalist footwear may promote foot and ankle strengthening and softer landings. Youth runners who currently wear traditional cushioned shoes and wish to transition to minimalist footwear should perform additional strengthening; the process of strengthening the foot and ankle and adapting to regular use of minimalist shoes can take many months. Promoting barefoot walking and use of minimalist footwear outside of sport may enhance the process.

CONCLUSION

Youth runners have specific needs that differ from adult runners. Accounting for age and maturity level can allow for safe participation in youth running and attaining the health benefits of the sport. Strength and conditioning professionals and others who work with youth runners should discourage early sport specialization in youth running. Participation in sports that involve jumping and multidirectional loading may promote skeletal health in youth runners. Psychological support of youth runners is important to avoid injury, overtraining, or burnout. Times of more rapid growth including puberty may result in youth runners being more susceptible to injury. Promoting proper nutritional intake and screening for disordered eating and menstrual dysfunction is critical to the health of runners. Determining safe distances to compete requires evaluating the maturity of the runner; special precautions are required to complete longer distances such as the half and full marathon. Foot strengthening and improving landing mechanics may help promote musculoskeletal development. Our recommendations are primarily expert-based opinion, and further research to develop evidence-based guidelines in youth runners is required.

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